Expt 7	**Spectral Estimation**
Date:	**7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal Signals Using Periodogram Method in MATLAB**

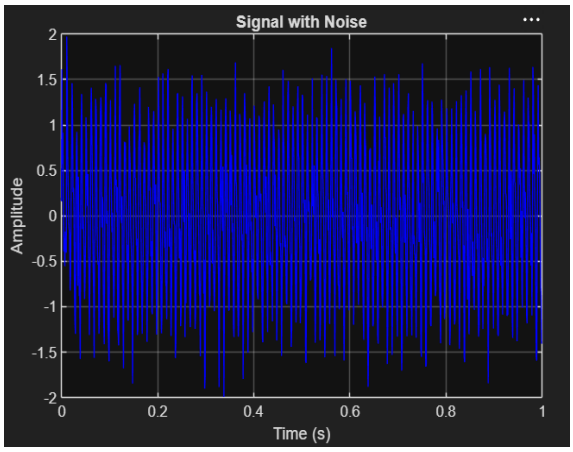
Code:

```
clc;
clear;
close all;

%% Signal
fs = 1000;
t = (0:999)/fs;
s = sin(2*pi*100*t) + 0.5*sin(2*pi*200*t);   % Original signal
n = 0.3*randn(size(t));                      % Noise
x = s + n;                                   % Noisy signal

%% Plot Signal with Noise
figure;
plot(t, x, 'b');
grid on;
title('Signal with Noise');
xlabel('Time (s)');
ylabel('Amplitude');

%% PSD using Periodogram
figure;
periodogram(x, [], [], fs);
title('PSD using Periodogram');

%% Magnitude Spectrum using FFT
```

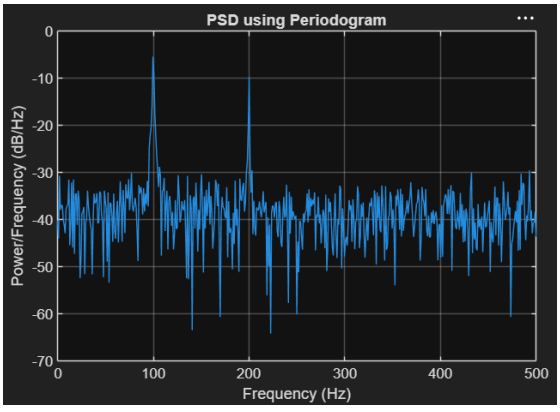
```
X = abs(fft(x));
f = (0:length(X)-1) * fs / length(X);
figure;
plot(f(1:500), X(1:500), 'LineWidth', 1.2);
grid on;
title('Magnitude Spectrum');
xlabel('Frequency (Hz)');
ylabel('|X(f)|');

%% Power Comparison (Signal vs Noise)
figure;
bar([mean(s.^2) var(n)]);
set(gca, 'XTickLabel', {'Signal','Noise'});
grid on;
title('Power Comparison');
ylabel('Power');
```

Output:

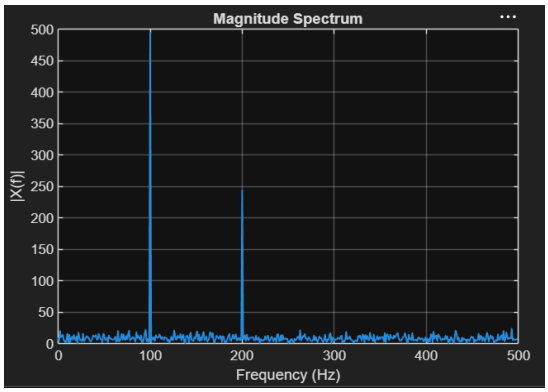
Observation 1:



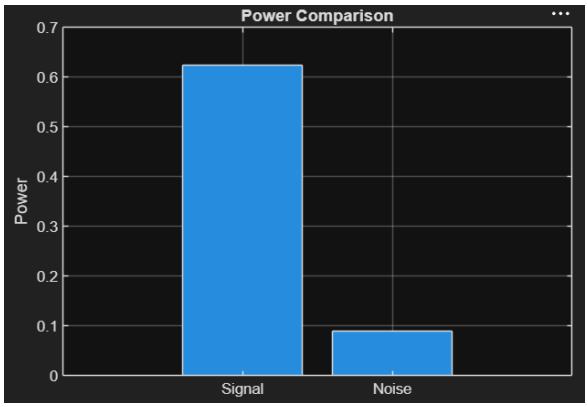
Observation 2:



Observation 3:



Observation 4:



Expt 7	**Spectral Estimation**
Date:	**7.2 Analysis of Windowing Effects on Spectral Estimation using Rectangular , Hamming and Hanning Windows in MATLAB**

Code:

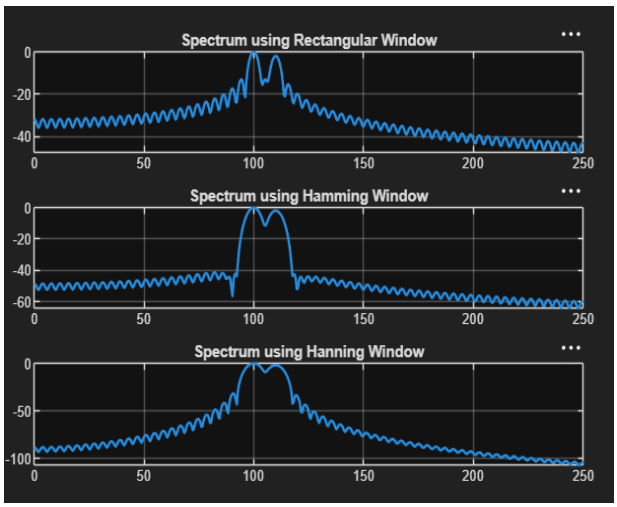
```
clc; clear; close all;


% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);


% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};


% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
```

```
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});

end

%% Individual Spectrums
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}),'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlim([0 250]); grid on;
end

%% Comparison
figure;
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;

%% Window Shapes
figure;
for i = 1:3
    subplot(3,1,i); plot(w{i},'LineWidth',1.5);
    title([names{i} ' Window']); grid on;
end

%% Peak Detection

for i = 1:3
    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
    fprintf('\n%s Window Frequencies:\n',names{i});
```

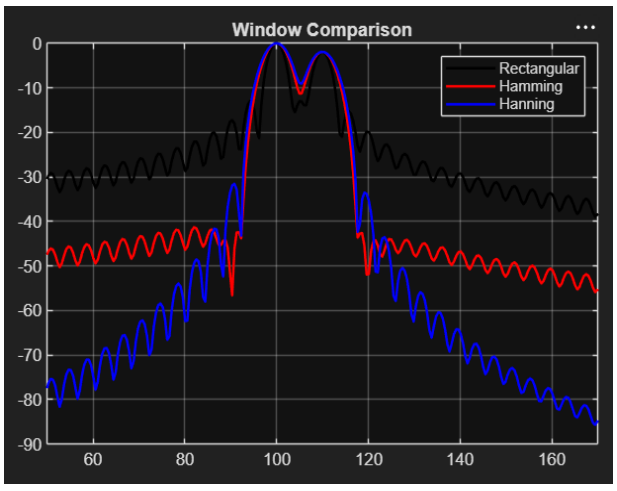
```
    disp(loc{i});
end

%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
    [~,idx] = max(P{i});
    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

Output:

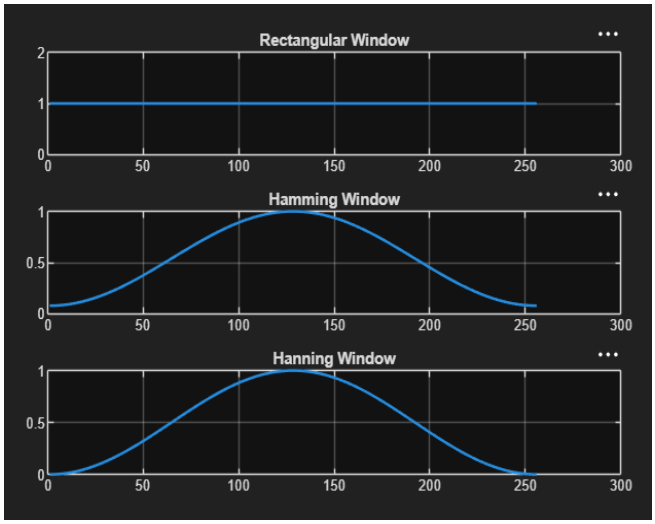
Objective 1:



Objective 2:



Objective 3:



Objective 4: